# Arista Access Node Optical Transport Failure Resolution Procedure for MAR-ACC-49

**Node:** MAR-ACC-49 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_1137

## Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) at the Marathahalli Core site, specifically on the MAR-ACC-49 node, is a critical failure domain that necessitates immediate attention due to its impact on the network's overall availability and performance. This failure domain occurs in the access role of the Arista equipment, which is responsible for providing connectivity to end-users and aggregating traffic from various sources. The underlying protocol mechanics, such as the Ethernet protocol's reliance on physical layer integrity, and hardware mechanics, including the optical transceivers' (SFP/XFP) susceptibility to degradation or damage, contribute to this failure domain. The Arista CLI dialect is utilized to diagnose and resolve this issue, leveraging commands that interact with the device's control and data planes to isolate and rectify the problem.

## Pre-requisite Checks

- Dispatch remote hands to Marathahalli Core site for physical inspection and potential intervention
- Verify redundant link status to ensure no single point of failure
- Check environmental conditions (temperature, humidity) at the site

## Potential Root Causes

| Cause | Probability |
|---|---|
| SFP/XFP degradation due to aging or environmental factors | High |
| Fiber cuts or damage caused by human error or environmental factors | Medium |
| Rx/Tx power loss due to laser or photodiode failure | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show interfaces transceiver details, Expected Output: Output displaying transceiver information, including optical power levels, Description: This command isolates issues related to the optical transceivers, such as SFP/XFP degradation, by displaying detailed information about the transceivers, including the optical power levels, which can indicate potential problems.

Command: show interfaces counters, Expected Output: Output displaying interface counters, including error and discard statistics, Description: This command provides insight into the interface's performance, including error and discard statistics, which can help identify issues related to the optical transport failure, such as high error rates or discards.
```

## Resolution Steps

Step 1: Step: Replace faulty SFP/XFP, Command: interface et1 replace transceiver, Description: This step involves physically replacing the faulty SFP/XFP with a new one, which requires careful handling to avoid damaging the surrounding equipment. The Arista CLI command 'interface et1 replace transceiver' is used to initiate the replacement process.

Step 2: Step: Adjust Rx/Tx power levels, Command: interface et1 transceiver power adjust, Description: This step involves adjusting the Rx/Tx power levels to optimal values, which can help resolve issues related to power loss. The Arista CLI command 'interface et1 transceiver power adjust' is used to adjust the power levels.

## Post-Check Verification

```
Command: show interfaces transceiver details, Expected Output: Output displaying updated transceiver information, including optical power levels, Description: This command verifies that the replacement or adjustment of the SFP/XFP has resolved the issue by displaying updated transceiver information, including optical power levels.

Command: show interfaces counters, Expected Output: Output displaying updated interface counters, including error and discard statistics, Description: This command verifies that the issue has been resolved by displaying updated interface counters, including error and discard statistics, which should indicate improved performance.
```

## Roll-back Procedures

- Step: Revert to original SFP/XFP, Command: interface et1 replace transceiver original, Description: If the replacement SFP/XFP does not resolve the issue, this step involves reverting to the original SFP/XFP, which requires careful handling to avoid damaging the surrounding equipment. The Arista CLI command 'interface et1 replace transceiver original' is used to initiate the reversion process.

- Step: Reset Rx/Tx power levels to default, Command: interface et1 transceiver power reset, Description: If the adjustment of Rx/Tx power levels does not resolve the issue, this step involves resetting the power levels to their default values. The Arista CLI command 'interface et1 transceiver power reset' is used to reset the power levels.